Governing Equations for Model 500 and 501

I. THE MODEL 500

In the tokamak coordinates (R, Z, ϕ), the magnetic field and velocity field can be expressed as follows

$$\mathbf{B} = F_0 \nabla\phi + \nabla\psi \times \nabla\phi, \tag{1}$$

$$\mathbf{v} = v_\parallel \mathbf{B} - R^2 \nabla u \times \nabla\phi. \tag{2}$$

Here, F_0 is the toroidal magnetic field and is approximately seen as constant in our study, while ψ is the poloidal magnetic flux. Further, u is the flow function for the $\mathbf{E} \times \mathbf{B}$ flow, $v_\parallel$ is the parallel velocity scaled by the magnetic field. And the governing equations are then:

$$\frac{\partial\psi}{\partial t} = \eta\left(T_e\right)\Delta^*\psi - R\left\{u, \psi\right\} - F_0\frac{\partial u}{\partial\phi}, \tag{3}$$

$$j = \Delta^*\psi, \quad j_\phi = -j/R, \tag{4}$$

$$\begin{aligned} R\nabla\cdot\left(R^2\rho\nabla_{pol}\frac{\partial u}{\partial t}\right) = {} & \frac{1}{2}\left\{R^2\left|\nabla_{pol}u\right|^2, R^2\rho\right\} + \left\{R^4\rho\omega, u\right\} + \left\{\psi, j\right\} - \frac{F_0}{R}\frac{\partial j}{\partial\phi} + \left\{\rho T_e, R^2\right\} \\ & + R\mu\left(T_e\right)\nabla^2\omega - \nabla\cdot\left[\left(\rho\rho_n S_{ion}\left(T_e\right) - \rho^2\alpha_{rec}\left(T_e\right)\right)R^2\nabla_{pol}u\right], \end{aligned} \tag{5}$$

$$\omega = \frac{1}{R}\frac{\partial}{\partial R}\left(R\frac{\partial u}{\partial R}\right) + \frac{\partial^2 u}{\partial Z^2}, \tag{6}$$

$$\frac{\partial\rho}{\partial t} = -\nabla\cdot\left(\rho\mathbf{v}\right) + \nabla\cdot\left(D_\perp\nabla_\perp\rho + D_\parallel\nabla_\parallel\rho\right) + \rho\rho_n S_{ion}\left(T_e\right) - \rho^2\alpha_{rec}\left(T_e\right), \tag{7}$$

$$\begin{aligned} \frac{\partial\left(\rho T_e\right)}{\partial t} = {} & -\mathbf{v}\cdot\nabla\left(\rho T_e\right) - \gamma\rho T_e\nabla\cdot\mathbf{v} + \frac{2}{3R^2}\eta\left(T_e\right)j^2 + \nabla\cdot\left(\kappa_\perp\nabla_\perp T_e + \kappa_\parallel\nabla_\parallel T_e\right) \\ & - \xi_{ion}\rho\rho_n S_{ion}\left(T_e\right) - \rho\rho_n P_L\left(T_e\right) - \rho^2 P_B\left(T_e\right), \end{aligned} \tag{8}$$

$$\begin{aligned} \rho B^2\frac{\partial v_\parallel}{\partial t} = {} & -\rho\frac{F_0}{2R^2}\frac{\partial}{\partial\phi}\left(B^2 v_\parallel^2\right) - \frac{\rho}{2R}\left\{B^2 v_\parallel^2, \psi\right\} - \frac{F_0}{R^2}\frac{\partial\left(\rho T_e\right)}{\partial\phi} + \frac{1}{R}\left\{\psi, \rho T_e\right\} \\ & + B^2\mu_\parallel\left(T_e\right)\nabla_{pol}^2 v_\parallel + \left(\rho^2\alpha_{rec}\left(T_e\right) - \rho\rho_n S_{ion}\left(T_e\right)\right)B^2 v_\parallel, \end{aligned} \tag{9}$$

$$\frac{\partial\rho_n}{\partial t} = \nabla\cdot\left(\mathbf{D}_n\cdot\nabla\rho_n\right) - \rho\rho_n S_{ion}\left(T_e\right) + \rho^2\alpha_{rec}\left(T_e\right) + S_n. \tag{10}$$

In the above equations, Eq. (3) is the induction equation, Eq. (4) is the result of the Ampère's law, Eq. (5) is the vorticity equation, while Eq. (6) is the definition of vorticity. Moreover, Eq. (7) is the continuity equation, Eq. (8) is the pressure equation, Eq. (9) is the parallel momentum equation, and finally Eq. (10) is the diffusive neutral species density equation. Here, we have assumed $T_e = T_i$, we also used $\Delta^* \psi \equiv R^2 \nabla \cdot (R^{-2} \nabla \psi)$, and the dissipative coefficient $\eta \propto T_e^{-3/2}$ and $\mu \propto T_e^{-3/2}$ are the resistivity and viscosity, while $\kappa_\perp$ and $\kappa_\parallel \propto T_e^{5/2}$ is the perpendicular heat conduction and parallel Braginskii heat conduction respectively. The parallel and perpendicular plasma diffusion coefficients and the neutral diffusion coefficient are $D_\parallel$, $D_\perp$ and $\mathbf{D}_n$ respectively, but we used $D_\parallel = 0 m^2/s$ in our study, so that the parallel density relaxation is purely carried out by convective flows. As for the coefficients governing the interaction between the plasmas and the neutrals, $S_{ion}(T)$ is the ionization rate and $\alpha_{rec}(T)$ is the recombination coefficient, the detailed form of which are described in Ref. [1]. Further, ξ_{ion} is the normalized deuterium ionization energy, $P_L(T_e)$ is the neutral line radiation coefficient and $P_B(T_e)$ the bremsstrahlung radiation[1]. The $\{f, g\}$ in the above equations denotes the Poisson bracket with $\{f, g\} \equiv R(\nabla f \times \nabla g) \cdot \nabla \phi$. Thus Eq. (3)-Eq. (10) form our governing equations.

II. THE MODEL 501

In model 501, we change the ion density to be the total density, and instead of the neutral density we now track the impurity species density with all the charged states included (including neutrals). Also, we write the momentum equation to represent the real momentum equation rather than the time derivative of the velocity field. Note that in model 501, we define the pressure as the following

$$P \equiv (\rho + \alpha_{imp} \rho_{imp}) T, \tag{11}$$

with $\alpha \equiv -1 + \left(\langle Z \rangle_{imp} + 1 \right) m_D / 2 m_{imp}$, and $T = 2 T_e$.

A. The Ampere's Law and the Induction Equation

In this case, the induction equation and the Ampere's Law is unchanged

$$\frac{\partial \psi}{\partial t} = \eta(T_e) \Delta^* \psi - R\{u, \psi\} - F_0 \frac{\partial u}{\partial \phi}, \tag{12}$$

$$j = \Delta^* \psi, \quad j_\phi = -j/R. \tag{13}$$

B. The Vorticity equation

The momentum equation (before written into reduced form) have the following form

$$\frac{\partial}{\partial t}(\rho \mathbf{v}) = \mathbf{J} \times \mathbf{B} - \nabla P + \nu \nabla_{pol}^2 \mathbf{v} - \nabla \cdot (\rho \mathbf{v}) \mathbf{v} - \rho \mathbf{v} \cdot \nabla \mathbf{v}. \tag{14}$$

We then apply $\nabla\phi \cdot \nabla \times (R^2 \cdots)$ on both side of the equation, on the RHS, we have

$$\begin{aligned}
\nabla\phi \cdot \nabla \times \left(R^2 \mathbf{J} \times \mathbf{B}\right) &= \nabla \cdot \left[R^2 (\nabla\phi \cdot \mathbf{J}) \mathbf{B} - R^2 (\nabla\phi \cdot \mathbf{B}) \mathbf{J}\right] \\
&= -\mathbf{B} \cdot \nabla j \\
&= -\frac{F_0}{R^2} \frac{\partial j}{\partial \phi} + (\nabla\psi \times \nabla j) \cdot \nabla\phi,
\end{aligned} \tag{15}$$

$$-\nabla\phi \cdot \nabla \times \left(R^2 \nabla P\right) = \nabla P \times \nabla R^2 \cdot \nabla\phi, \tag{16}$$

$$\nabla\phi \cdot \nabla \times \left(R^2 \nu \nabla_\perp^2 \mathbf{v}\right) = \mu \nabla_\perp^2 \omega, \tag{17}$$

$$\begin{aligned}
-\nabla\phi \cdot \nabla \times \left[R^2 \rho \mathbf{v} \cdot \nabla (\mathbf{v})\right] &= -\nabla\phi \cdot \nabla \times \left[\frac{1}{2} R^2 \rho \nabla v_{pol}^2 - R^2 \rho \mathbf{v}_{pol} \times (\nabla \times \mathbf{v}_{pol})\right] \\
&= -\nabla\phi \cdot \left[\frac{1}{2} \nabla \left(R^2 \rho\right) \times \nabla v_{pol}^2\right] \\
&\quad + \nabla\phi \cdot \nabla \times \left[R^2 \rho \mathbf{v}_{pol} \times (\nabla \times \mathbf{v}_{pol})\right] \\
&= -\nabla\phi \cdot \left[\frac{1}{2} \nabla \left(R^2 \rho\right) \times \nabla v_{pol}^2\right] \\
&\quad + \nabla\phi \cdot \left[\nabla \left(\rho \omega R^4\right) \times \nabla_{pol} u\right],
\end{aligned} \tag{18}$$

$$\begin{aligned}
-\nabla\phi \cdot \nabla \times \left[R^2 \mathbf{v} \nabla \cdot (\rho \mathbf{v})\right] &= -\nabla \cdot \left[\phi \nabla \times R^2 \mathbf{v} \nabla \cdot (\rho \mathbf{v})\right] \\
&= \nabla \cdot \left[\nabla\phi \times R^2 \mathbf{v} \nabla \cdot (\rho \mathbf{v})\right] \\
&= -\nabla \cdot \left[R^2 \nabla_{pol} u \nabla \cdot (\rho \mathbf{v})\right].
\end{aligned} \tag{19}$$

Thus we finally get the vorticity equation

$$\begin{aligned}
R\nabla \cdot \left[R^2 \left(\rho \nabla_{pol} \frac{\partial u}{\partial t} + \nabla_{pol} u \frac{\partial \rho}{\partial t}\right)\right] &= \frac{1}{2} \left\{R^2 \left|\nabla_{pol} u\right|^2, R^2 \rho\right\} + \left\{R^4 \rho \omega, u\right\} \\
&\quad - R\nabla \cdot \left[R^2 \nabla_{pol} u \nabla \cdot (\rho \mathbf{v})\right] + \{\psi, j\} - \frac{F_0}{R} \frac{\partial j}{\partial \phi} \\
&\quad + \left\{P, R^2\right\} + R\mu (T_e) \nabla_{pol}^2 \omega.
\end{aligned} \tag{20}$$

Here, we again have

$$\omega = \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial u}{\partial R}\right) + \frac{\partial^2 u}{\partial Z^2}. \tag{21}$$

C. The Total Continuity Equation

Then we move on to the density equation

$$\frac{\partial}{\partial t}\rho = -\nabla\cdot(\rho\mathbf{v}) + \nabla\cdot\left[D_D\nabla\left(\rho-\rho_{imp}\right)\right] + \nabla\cdot\left(D_{imp}\nabla\rho_{imp}\right) + S_D + S_{imp}. \tag{22}$$

Here, D_D and D_{imp} are the diffusion for deuterium and the impurity species respectively, and ρ_{imp} is the total density of impurity species.

D. The Pressure Equation

Meanwhile, the pressure equation is

$$\begin{aligned}\frac{\partial\left(P\right)}{\partial t} = &-\mathbf{v}\cdot\nabla\left(P\right) - \gamma P\nabla\cdot\mathbf{v} + \frac{\gamma-1}{R^2}\eta\left(T_e\right)j^2 + \nabla\cdot\left(\kappa_\perp\nabla_\perp T + \kappa_\parallel\nabla_\parallel T\right)\\ &-n_e n_{imp}P_{rad}\left(T_e\right) + \frac{\gamma-1}{2}\mathbf{v}\cdot\mathbf{v}\left(S_D+S_{imp}\right).\end{aligned} \tag{23}$$

Here, n_e is calculated via the coronal equilibrium assumption, and $P_{rad}\left(T_e\right)$ is the radiation function assuming coronal equilibrium. The last term derives from the friction force, similar to the kinetic to thermal energy conversion during inelastic collision. In the code, we have

$$\mathbf{v}\cdot\mathbf{v} = v_\parallel^2 B^2 + R^2\left(\nabla_{pol}u\right)^2. \tag{24}$$

E. Parallel Velocity Equation

As for the parallel velocity equation, we time $\mathbf{B}$ on both side of the equation so that we have

$$\mathbf{B}\cdot\mathbf{J}\times\mathbf{B} = 0, \tag{25}$$

$$-\mathbf{B}\cdot\nabla P = -\frac{F_0}{R^2}\frac{\partial P}{\partial\phi} - \nabla\psi\times\nabla\phi\cdot\nabla P, \tag{26}$$

$$\nu\mathbf{B}\cdot\nabla_{pol}^2\mathbf{v} = B^2\mu_\parallel\left(T_e\right)\nabla_{pol}^2 v_\parallel, \tag{27}$$

$$-\mathbf{B}\cdot\mathbf{v}\nabla\cdot\left(\rho\mathbf{v}\right) = -B^2\nabla\cdot\left(\rho\mathbf{v}\right)v_\parallel, \tag{28}$$

$$\begin{aligned}-\mathbf{B}\cdot\left(\rho\mathbf{v}\cdot\nabla\mathbf{v}\right) &= -\frac{1}{2}\rho\mathbf{B}\cdot\nabla\left(v_\parallel B\right)^2\\ &= -\frac{1}{2}\rho\frac{F_0}{R^2}\frac{\partial}{\partial\phi}\left(v_\parallel B\right)^2 + \frac{1}{2}\rho\nabla\psi\times\nabla\left(v_\parallel B\right)^2\cdot\nabla\phi.\end{aligned} \tag{29}$$

Hence the parallel momentum equation can finally be written as

$$B^2\frac{\partial}{\partial t}\left(\rho v_\parallel\right) = -\frac{1}{2}\rho\frac{F_0}{R^2}\frac{\partial}{\partial\phi}\left(v_\parallel B\right)^2 - \frac{\rho}{2R}\left\{B^2 v_\parallel^2, \psi\right\} - \frac{F_0}{R^2}\frac{\partial P}{\partial\phi} + \frac{1}{R}\left\{\psi, P\right\}$$
$$-B^2\nabla\cdot\left(\rho\mathbf{v}\right)v_\parallel + B^2\mu_\parallel\left(T_e\right)\nabla_{pol}^2 v_\parallel. \tag{30}$$

F. Impurity Continuity Equation

And ultimately, we have the continuity equation for the impurity species

$$\frac{\partial}{\partial t}\rho_{imp} = -\nabla\cdot\left(\rho_{imp}\mathbf{v}\right) + \nabla\cdot\left(D_{imp}\nabla\rho_{imp}\right) + S_{imp}. \tag{31}$$

III. THE MODEL 501 WITH IONIZATION ENERGY

Optionally, we can include the ionization energy into the energy equation. Assuming coronal equilibrium, the ionization potential energy $E_{ion}\left(T_e, n_e\right)$ for average charge number $Z_{imp}\left(T_e, n_e\right)$ contribute an additional term to the equation of energy (neglecting deuterium contribution since it is small), the value of this ionization potential energy is $n_{imp}E_{ion}\left(n_e, T_e\right)$.

Instead of the previous equation of state

$$\frac{\partial P}{\partial t} + \nabla\cdot\left(\mathbf{v}P\right) = -\left(\gamma-1\right)P\nabla\cdot\mathbf{v} + \frac{\gamma-1}{R^2}\eta\left(T_e\right)j^2 + \nabla\cdot\left(\kappa_\perp\nabla_\perp T + \kappa_\parallel\nabla_\parallel T\right)$$
$$-n_e n_{imp}P_{rad}\left(T_e\right) + \frac{\gamma-1}{2}\mathbf{v}\cdot\mathbf{v}\left(S_D + S_{imp}\right). \tag{32}$$

Define $P^* \equiv P + (\gamma-1)n_{imp}E_{ion}$. The new equation of state is now

$$\frac{\partial P^*}{\partial t} + \nabla\cdot\left(\mathbf{v}P^*\right) = -\left(\gamma-1\right)P\nabla\cdot\mathbf{v} + \frac{\gamma-1}{R^2}\eta\left(T_e\right)j^2 + \nabla\cdot\left(\kappa_\perp\nabla_\perp T + \kappa_\parallel\nabla_\parallel T\right)$$
$$+\nabla\cdot\left[\left(\gamma-1\right)E_{ion}D_{imp}\nabla n_{imp}\right]$$
$$-n_e n_{imp}P_{rad}\left(T_e\right) + \frac{\gamma-1}{2}\mathbf{v}\cdot\mathbf{v}\left(S_D + S_{imp}\right). \tag{33}$$

The idea of this form of equation is that the summation of the thermal energy and the ionization potential energy should conserve, part from other source & sinks. The conductive thermal energy flux and the diffusive ionization energy flux are moved to the third and fourth term at the RHS respectively. Here, the ionization potential energy is not considered lost when the impurities recombine to avoid double counting since we already have a recombination radiation component in our radiation function.

So that it now explicitly has the form

$$\begin{aligned}\frac{\partial(P)}{\partial t} = & -\mathbf{v}\cdot\nabla(P) - \gamma P\nabla\cdot\mathbf{v} + \frac{2}{3R^2}\eta(T_e)\,j^2 + \nabla\cdot\left(\kappa_\perp\nabla_\perp T + \kappa_\parallel\nabla_\parallel T\right) \\ & +\nabla\cdot\left[(\gamma-1)\,E_{ion}D_{imp}\nabla n_{imp}\right] \\ & -\frac{2}{3}n_{imp}\frac{\partial E_{ion}}{\partial T_e}\frac{\partial T_e}{\partial t} - n_e n_{imp}P_{rad}(T_e) - \frac{2}{3}E_{ion}\frac{\partial n_{imp}}{\partial t} - \frac{2}{3}E_{ion}n_{imp}\nabla\cdot\mathbf{v} \\ & -\frac{2}{3}\left[n_{imp}\frac{\partial E_{ion}}{\partial T_e}\mathbf{v}\cdot\nabla T_e + E_{ion}\mathbf{v}\cdot\nabla n_{imp}\right] + \frac{\gamma-1}{2}\mathbf{v}\cdot\mathbf{v}\,(S_D + S_{imp})\,. \end{aligned} \tag{34}$$

Meanwhile, we have

$$E_{ion}(n_e, T_e) \equiv \sum_i P_i(n_e, T_e)\,E_i, \quad \partial_t E_{ion} = \sum_i \partial_t P_i(n_e, T_e)\,E_i, \tag{35}$$

with E_i being the accumulated ionization energy of charge state i. Essentially the inclusion of the ionization potential energy contribution add three amat terms to the LHS of the equation and two additional terms to the RHS. **Note we have ignored all $\partial E_{ion}/\partial n_e$ terms as those are small for the parameter regime we are considering.**

The above treatment has the advantage that it would not cause any artificial recombination radiation due to the instant recombination under Coronal Equilibrium assumption. On the other hand, it is equivalent to assuming a strong three-body recombination (since we feed the ionization energy back into the thermal energy) while there is none. This is the inherent limitation of the Coronal Equilibrium assumption. To fix this one would need to go for a future model with self-consistent evolution of ionization states.

IV. HEATING BY VISCOUS TERMS

If we are to add the heating by viscous terms, in principle, we should take the momentum equation, time both side with the conjugation of the velocity then the resulting term which corresponded with the viscous term should be our viscous heating.

For code implementation, the perpendicular viscous term is a bit more complicated since the perpendicular equation is written in the second derivative of the flow function (thus we should time the viscous term with the conjugation of the flow function). On the other hand, the parallel viscosity is more relevant to the physics we are concerned (ion heating by Pavel etc.). So we would begin with the parallel viscous heating.

The kinetic energy depletion by such viscous damping is simply:

$$\begin{aligned} P_\mu &= v_\parallel \mu_\parallel B \nabla^2_{pol} \left(v_\parallel B \right) \\ &= \nabla_{pol} \cdot \left[\mu_\parallel v_\parallel B \nabla_{pol} \left(v_\parallel B \right) \right] - \mu_\parallel \left[\nabla_{pol} \left(v_\parallel B \right) \right]^2 . \end{aligned} \tag{36}$$

The first term on the RHS is a viscous kinetic energy transport flux, and the second term on the RHS is the conversion of the kinetic energy into thermal energy. Thus the corresponding heat source in the pressure equation should then simply be $\delta \partial_t P = (\gamma - 1) \mu_\parallel \left[\nabla_{pol} \left(v_\parallel B \right) \right]^2$.

V. THE EFFECTIVE CHARGE IN THE RESISTIVITY

The effective charge is such a pain.

The definition of effective charge as follows:

$$Z_{eff} \equiv \frac{\sum_i n_i Z_i^2}{\sum_i n_i Z_i}. \tag{37}$$

In JOREK model 501, this means:

$$Z_{eff} = \frac{\rho + (\mu_{imp} \sum_i P_{imp}(i) Z_i^2 - 1) \rho_{imp}}{\rho + \beta_{imp} \rho_{imp}}. \tag{38}$$

Here $P_{imp}(i)$ is the distribution of charge states, Z_i their respective charge, μ_{imp} is the mass ratio between the background ion and the impurity, and $\beta_{imp} \equiv \mu_{imp} Z_{imp} - 1$. The temperature gradient is then

$$\begin{aligned} \frac{\partial Z_{eff}}{\partial T_e} = & \frac{\mu_{imp} \sum_i \partial P_{imp}(i) / \partial T_e Z_i^2 \rho_{imp}}{\rho + \beta_{imp} \rho_{imp}} \\ & - \frac{\rho + (\mu_{imp} \sum_i P_{imp}(i) Z_i^2 - 1) \rho_{imp}}{(\rho + \beta_{imp} \rho_{imp})^2} \frac{\partial \beta_{imp}}{\partial T_e} \rho_{imp}. \end{aligned} \tag{39}$$

Then the total density gradient is

$$\frac{\partial Z_{eff}}{\partial \rho} = \frac{1}{\rho + \beta_{imp} \rho_{imp}} - \frac{\rho + (\mu_{imp} \sum_i P_{imp}(i) Z_i^2 - 1) \rho_{imp}}{(\rho + \beta_{imp} \rho_{imp})^2}. \tag{40}$$

Last the impurity density gradient is

$$\frac{\partial Z_{eff}}{\partial \rho_{imp}} = \frac{\mu_{imp} \sum_i P_{imp}(i) Z_i^2 - 1}{\rho + \beta_{imp} \rho_{imp}} - \frac{\rho + (\mu_{imp} \sum_i P_{imp}(i) Z_i^2 - 1) \rho_{imp}}{(\rho + \beta_{imp} \rho_{imp})^2} \beta_{imp}. \tag{41}$$

We have ignored any density dependence in P_{imp} for simplicity and their insignificance. The Spitzer resistivity indicates[2]:

$$\eta = \frac{\sqrt{2 m_e} Z_{eff} e^2 \ln \Lambda}{12 \pi^{3/2} \epsilon_0^2 T_e^{3/2}} \times F(Z_{eff}), \quad F(Z_{eff}) = \frac{1 + 1.198 Z + 0.222 Z^2}{1 + 2.966 Z + 0.753 Z^2}. \tag{42}$$

VI. PAUL PARKS'S ABLATION SCALING!

Our friend Paul Parks kindly pointed out the Sergeev scaling we are using now tends to overestimate the ablation rate by a factor of 2 compared with his model, and he argues that his result in more accurate. Also, he pointed out that in the neon/D2 mixture case the neon and deuterium mixed together to form the pellet, thus treating them as formed separately is not appropriate, while for argon/D2 pellet it is appropriate.

In the following, I will try to outline the new mixed pellet ablation treatment we will be using.

A. Neon/Deuterium mixed pellet

We define the **molecular** mixture ratio

$$X \equiv \frac{N_{D2}}{N_{Ne} + N_{D2}}. \tag{43}$$

Meanwhile the **atom number** mixture ratio has the following relationship:

$$Y \equiv \frac{N_D}{N_{Ne} + N_D}, \quad X = \frac{Y}{2 - Y}, \; Y = \frac{2X}{X + 1}. \tag{44}$$

The mass ablation rate is shown to have the following scaling law in Paul's model

$$G\left[g/s\right] = \lambda\left[X\right] \left(\frac{T_e\left[eV\right]}{2000}\right)^{5/3} \left(\frac{r_p\left[cm\right]}{0.2}\right)^{4/3} n_e\left[10^{14} cm^{-3}\right]^{1/3}, \tag{45}$$

$$\lambda\left[X\right] \equiv 27.0837 + \tan\left(1.48709X\right). \tag{46}$$

The ablation rate for each species is then

$$\frac{d}{dt} N_{Ne}\left[mol/s\right] = \frac{\left(1 - X\right) G}{W_{Ne}\left(1 - X\right) + X W_{D2}}, \tag{47}$$

$$\frac{d}{dt} N_{D2}\left[mol/s\right] = \frac{XG}{W_{Ne}\left(1 - X\right) + X W_{D2}}, \tag{48}$$

where we have

$$W_{Ne}\left[g/mol\right] = 20.183, \quad W_{D2}\left[g/mol\right] = 4.0282. \tag{49}$$

And the pellet surface recession speed is then

$$\frac{d}{dt}\left(r_p^{5/3}\right) = -3.572 \times 10^{-6} \frac{\lambda\left(X\right)}{\rho_0\left(X\right)} T_e\left[eV\right]^{5/3} n_e\left[10^{14} cm^{-3}\right]^{1/3}. \tag{50}$$

Here, the mass density of the mixed pellet should be

$$\rho_0(X) = \left[1 - X + \frac{W_{D2}}{W_{Ne}}X\right]\left[\frac{1-X}{\rho_{Ne}} + \frac{W_{D2}}{W_{Ne}}\frac{X}{\rho_{D2}}\right]^{-1}. \tag{51}$$

In terms of **molecular** number density we have

$$n_m(X) = \left[\frac{X}{n_{D2}} + \frac{1-X}{n_{Ne}}\right]^{-1}. \tag{52}$$

Meanwhile, in terms of atom number density we have

$$n_A = (1+X)\left[\frac{X}{n_{D2}} + \frac{1-X}{n_{Ne}}\right]^{-1} = \left[\frac{Y}{n_D} + \frac{1-Y}{n_{Ne}}\right]^{-1}. \tag{53}$$

B. Argon/Deuterium mixed pellet

For argon and deuterium mixed pellet, the argon part and the deuterium part are separate, and their ablation should be treated separately as we were doing in the older version. Following Paul Parks model, the pure argon fragment ablation rate would be

$$G\,[g/s] = \lambda\left(\frac{T_e\,[eV]}{2000}\right)^{5/3}\left(\frac{r_p\,[cm]}{0.2}\right)^{4/3} n_e\left[10^{14}cm^{-3}\right]^{1/3}, \quad \lambda \equiv 36.6337. \tag{54}$$

Meanwhile the pure deuterium fragment ablation rate would be

$$G\,[g/s] = \lambda\left(\frac{T_e\,[eV]}{2000}\right)^{5/3}\left(\frac{r_p\,[cm]}{0.2}\right)^{4/3} n_e\left[10^{14}cm^{-3}\right]^{1/3}, \quad \lambda \equiv 39.0023. \tag{55}$$

C. Dependence on the B field

A preliminary scaling law with B field dependence can be given by replacing the λ above with

$$\lambda_{new} = \lambda_{old} * (\frac{2}{B\,[T]})^{\nu}, \quad \nu = 0.843. \tag{56}$$

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